

Stability Lens / GeoStab-FT: final Serova build plan

Research and implementation blueprint

Source: HACKATHON_PLAN.md (working copy)

SHA-256: 9fae3079f6244144bab066b36a2088d235ca868529874ae02fc044d531611c52

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Primary foundation model: facebook/esm2_t30_150M_UR50D, adapted with rank-8 LoRA on key/value projections. **3D foundation model:** Boltz-2, kept frozen as a coordinate generator. **Supervised data:** the supplied Serova peptide–HLA half-life CSV. **Workspace:** this repository on sh-arka22/are.

Status: final implementation blueprint, not a trained model or a measured accuracy claim. Existing split tools and data remain; the model/structure pipeline and its tests still need implementation. This revision changes only this local master plan and does not launch compute. Modal access and sponsored credits are user-confirmed, but a numerical spend cap must be approved before paid jobs.

This master plan governs the current architecture and supersedes conflicting older model priorities in the unchanged README and Science Skills companion. Their supporting biological/evidence context remains useful. The pre-event primer remains background, not event-built implementation.

0. Decision and alignment with the Serova track

Use the official Serova Protein Engineering Track Challenge as the task specification. It names ESM-2, Boltz-2, and ProteinMPNN, supplies the half-life dataset, and explicitly permits fine-tuning and prediction heads. The event page supplies context; the 25-page build proposal is assessed critically rather than treated as established science.

The core experiment is a controlled comparison: supervised baselines → frozen ESM-2 controls → genuine ESM-2 adaptation → a bounded, matched 3D extension. Train a residue-level peptide–HLA interaction head rather than relying only on mean-pooled embeddings. Evaluate on the existing frozen split, with train-only preprocessing, validation-only model selection, exact fallbacks, support counts, uncertainty, and measured cost.

The goal is better measured half-life prediction, not affinity prediction, attractive structures, or an arbitrary leaderboard target. Negative or inconclusive results remain valid if supported. No clinical, vaccine-efficacy, guaranteed-accuracy, or first-in-literature claim is implied.

Explicit Serova alignment

Serova requirement	Concrete response
§1: can existing protein foundation models improve complex stability prediction?	Compare X150 frozen and L150 adapted ESM-2 on measured half-life; then isolate Boltz-derived geometry
§3: named model families	ESM-2 and Boltz-2 are core; ProteinMPNN is a listed optional inverse-folding extension
§4: supplied dataset and target	Use the checked-in 28,166-row CSV and thalf_hours; preserve original allele/domain/pseudosequence meaning
§5: establish sensible splits and a supervised baseline	Reuse the frozen Hamming≤2 split, harden its checks, and include B0/B1
§5: fine-tune or build prediction heads	Train LoRA inside ESM-2 and a residue-interaction head; compare original-weight partial unfreezing only if budget permits

Serova requirement	Concrete response
§5: account for compute and allow subsetting	Profile first, enforce a numerical spending cap, and limit the structural cohort
§5: negative findings are valid	Report failed, negative, and inconclusive comparisons; never guarantee a score
§6: biological context and evidence	Distinguish kinetics/affinity, verify constructs, handle structures/failures explicitly, and report paired metrics with cost

Required build versus optional research

Core future build: correct data/evaluation loading; B0/B1; F150/X150 frozen controls; one genuine L150 adaptation run; a validated Boltz pilot and bounded G_conf/G_pair comparison; saved predictions, paired results, and a real-output demo.

Optional, after the core is viable: U150 partial unfreezing, the fixed contact-tensor comparator, repeated training seeds, hydrogen-bond/SASA descriptors, structural ensembles, ProteinMPNN, separate calibration, larger backbones, and external-data transfer.

F150 is a cheap pooled-head control sharing the X150 frozen encoder cache, not another backbone training campaign. Only L150 is the mandatory encoder-adaptation run. Do not let optional work delay the core pair of adaptation/3D experiments.

A sequence-only result can still be a legitimate Serova fallback submission, but it does **not** mean the user's requested 3D experiment was completed. Label an unrun structural branch as partial/unfulfilled; a completed negative structural comparison is a valid completed experiment.

1. Evidence and current repository state

What actually exists

Asset	Current state	Consequence
context/dataset.csv and .xlsx	Checked-in challenge data	Use the CSV as the canonical computational input; do not create an untracked competing export
splits/peptide_split.csv, allele_split.csv, split_report.json	Frozen assignments and saved generation report	Consume and fingerprint them; do not regenerate to improve a score
scripts/make_splits.py	Existing generator	Running it overwrites the split artifacts
scripts/audit_splits.py	Existing independent distance computation and diagnostics	Its current pass/fail gates do not fully enforce the current split contract
scripts/measure_leakage_by_distance.py	Local earlier diagnostic; newer upstream version inspected	Distinguish local behavior from upstream analysis A/B
experiments/	Proposals, template, and result placeholders	A proposal is not a trained model or a measured result
This plan, Science Skills companion, README	Current planning/documentation layer	Define future requirements, not completed scientific work

Asset	Current state	Consequence
Training code, model tests, evidence registries, structures, app	Not implemented in this checkout	Do not advertise runnable training/deployment commands or invented results

This existing repository is the project workspace; do not initialize another Git root under `submission/`. The [background primer](#) was written before the event and is retained as attributed reference material, not newly built hackathon implementation. Event eligibility and source redistribution conditions remain organizer/source questions, not assumptions inferred from public GitHub availability.[3][19]

- Read all extractable text from the three-page Serova brief and the 25-page proposal. Graphic-only illustrations were not treated as independently verified specifications.
- Confirmed the new Serova PDF is byte-identical to `context/serova_primer.pdf`.
- Inspected current Git state and tracked files: split tools and proposals exist; no model pipeline, packaged environment, or model test suite exists.
- Used Tavily extraction, bounded dynamic search, and a completed deep-research request; checked consequential recommendations against primary/official sources.
- Verified ESM-2 model/configuration identities, attention-module names, official Boltz options and output semantics, Biotite hydrogen requirements, and Modal public prices.
- Read primary protein-fine-tuning and PPI PEFT evidence. These establish feasibility on other tasks, not a half-life improvement here.
- Inventoried candidate structural-cohort membership using only allele, peptide, component, and partition metadata; no target values entered that inventory.
- Did not rerun scientific model metrics, download weights, install software, or use GPU compute.

Input identities

Input	SHA-256
Supplied Serova PDF and repository challenge PDF	44f0d7ef591e069bdb76b15c1597ce34603b055e355547554907de8f22d0bd33
Supplied 25-page build proposal	a0e212c91ec8565c86e436a63b6b9269f6a2961d2c6161541885fae3f6cab0bc
Canonical dataset CSV, retained identity	384e0accd35c589f02bd3f7d702d31b00a27f8f6f728cacb52b4d8a6c83af44a
Frozen primary split	0d92a33bb64eb35d922f698283ca08f6556019e55c7f87c7ee5d31ab6b4d7988

Evidence supporting adaptation

- Serova §5 explicitly permits fine-tuning, masking, different seeds, and prediction heads. The proposal PDF's blanket fine-tuning veto is not a challenge rule.
- The 2024 Nature Communications study reports fine-tuning benefits across protein tasks and resource benefits from PEFT. It also finds full fine-tuning of a smaller ESM-2 150M competitive in one task. Neither finding supplies a Serova score.
- The 2023 MLSB workshop PPI study demonstrates ESM-2 LoRA and favorable key/value adaptation on its task. It supports the starting design, not a universal best adapter configuration.

- ESM-2 150M is public, ungated, MIT-licensed, and not a released stability-trained checkpoint.
- Boltz-2 supplies complex coordinates/confidence; its affinity property is for a small-molecule ligand, not this protein peptide.

The generated deep-research report contained provider-specific limits and broad suggestions not supported for this exact workflow. Do not copy its universal token limits, fictitious chain concatenation, censoring assumptions, or replacement metrics without primary evidence.

2. Extracted scripts and validation gaps

Local versus inspected upstream

Script	Local Git blob	Upstream Git blob at de2532f	Status
make_splits.py	21e96d3bb92d8ed910a73c492b6855e601f61533	Same	Identical
audit_splits.py	7b967dcbde8d7d97c094c0f60a1236ad7dc2c2da	Same	Identical
measure_leakage_by_distance.py	3dc48a1615af803ca8fe1d0d9bde49eff855a630	2f3f39cebd88aebd0d246aa7b9d19200b623c6d4	Upstream adds A; local remains the earlier version

The upstream [reconciliation](#) and [brainstorm](#) are reviewed source material, not files imported by this revision. Their assertions must be checked against the code rather than adopted wholesale.[47][48]

Generator: `scripts/make_splits.py`

Input: canonical CSV. **Dependencies:** pandas and standard library. **Outputs overwritten when run:** the primary split CSV, secondary split CSV, and JSON report.[43]

- `UnionFind` and `cluster_peptides` (69–112) build exact global connected components using masked-position keys. At the active cutoff of two, groups include two-substitution neighbors as well as closer pairs.
- `greedy_assign` (146–172) assigns whole groups, largest first, according to relative remaining row quota with seeded tie-breaking. This avoids the old absolute-deficit imbalance; the primary allocation is row-weighted, not label-bin stratified.
- `greedy_assign_stratified` (128–143) applies that allocation inside secondary allele strata. `main` creates those strata from full-dataset per-allele median labels (290–297).
- `verify_no_cross_split_neighbours` (178–210) compares every nontraining peptide against training peptides. It does not separately check validation-versus-test distances.
- `validate_detector` (213–226) plants a one-substitution positive, not a two-substitution boundary case.
- `main` (257–394) reports cutoff sensitivity, validates several invariants, and writes fixed output paths. It asserts uniform peptide length, not specifically nine. The `if __name__ == "__main__"` guard prevents regeneration on import.

The module's one-edit convention description and historical threshold rationale are not the authoritative current constant: `CLUSTER_MAX_DIST = 2`. Generation is not an onboarding step or a harmless verification command.

Independent audit: `scripts/audit_splits.py`

Inputs: CSV and saved primary assignment. **Dependencies:** NumPy/pandas, with SciPy used by pandas Spearman calculations. **Outputs:** stdout and exit status; no scientific artifact writes.[44]

The audit joins on peptide with `validate="many_to_one"`, computes Hamming distances by chunked brute force, and checks train–validation, train–test, validation–test, several coverage invariants, duplicate pairs, memorization coverage, and distribution summaries. It reads evaluation labels for diagnostic calculations and executes at import time.

Its three distance pass/fail checks still reject only distances ≤ 1 (48–61). Printing the number of neighbors at distance ≤ 2 does not enforce the two-edit contract. `peptide_lookup_score` returns `0.0` when fewer than ten matches exist (76–85); this is a sentinel, not a measured zero correlation. The current checks also do not replace full schema validation, exact assignment-key equality, or a complete secondary-allele audit.

Distance diagnostic: inspected upstream `scripts/measure_leakage_by_distance.py`

Inputs: CSV and saved primary assignment. **Dependencies:** NumPy/pandas/SciPy. **Outputs:** printed descriptive statistics and a null-tolerance check. Both analyses execute at import/run time; no train-only CLI switch exists.[45]

The upstream file builds `Z[(peptide, allele)]` from `log1p` labels standardized per allele over the **entire dataset** (45–47). Analysis A later selects training peptide pairs (69–96) but uses that table; B compares test with training pairs (98–126).

Therefore A is not strictly train-only. An isolated synthetic check during this review held training labels fixed and changed only held-out labels; training normalized values and pooled ranks changed. This proves a dependency, not its numerical magnitude on the real dataset. Within a single finite, nondegenerate allele group, affine standardization preserves ranks; pooling multiple groups is different.

Other limits:

- Pair sampling takes at most three sorted-index neighbors per distance. A applies this cap before removing duplicate pair orientations, so eligible pairs can be missed (81–83).
- The A null uses the first 3,000 sorted training peptides, can sample self-matches, and couples RNG draws to set iteration. Determinism and representativeness need explicit policies.
- `report` prints a null correlation and tests `abs(null_rho) > 0.08` (54–66). An isolated empty-null fixture returned NaN without failing this guard.
- The final statement that the boundary lies between distances two and three is unconditional text (128–129), not a conclusion established by an implemented decision rule.

The source review and synthetic checks did not run the real dataset diagnostic, regenerate assignments, or reproduce its reported correlations. The newer A/B script remains upstream-only in this personal branch.

Required future hardening, not completed fixes

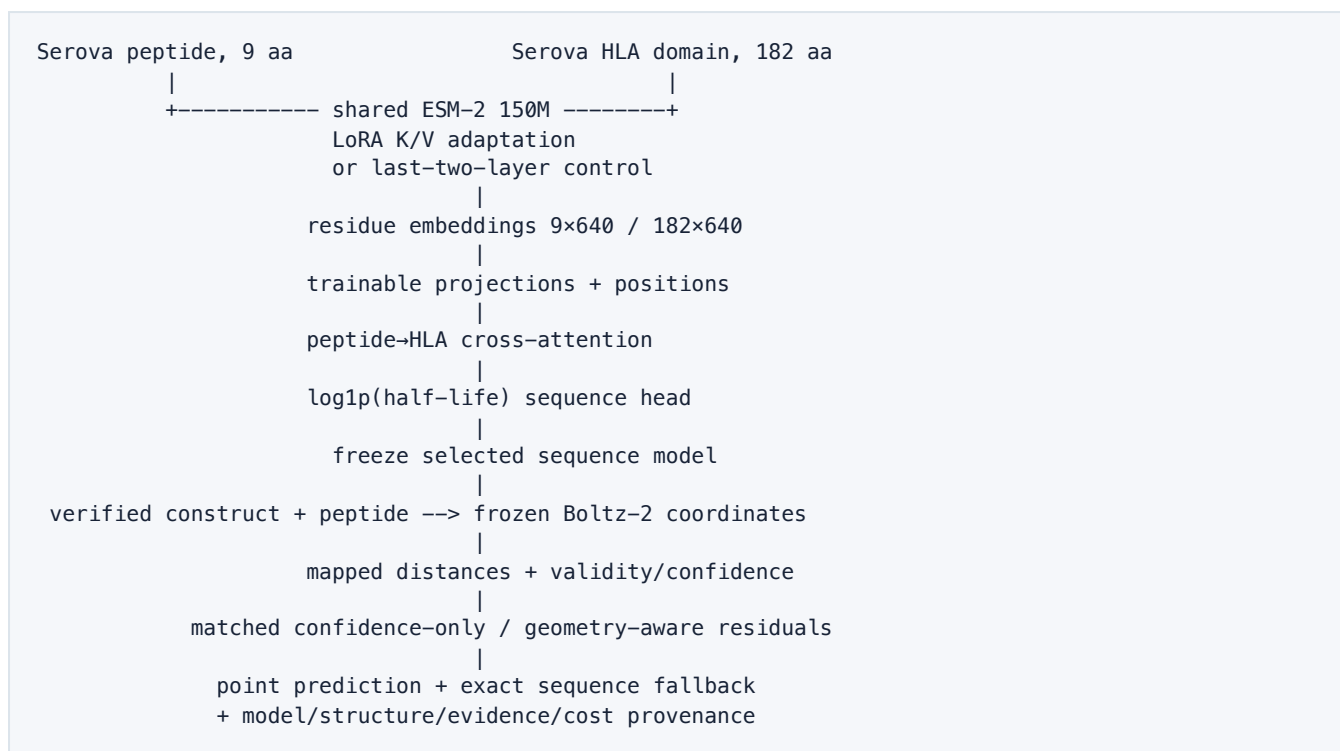
ID / priority	Gap	Required acceptance before relying on the result
D1 / P0	All-row normalization enters nominally train-only A	Fit preprocessing on training rows only; perturbing/removing held-out labels cannot change A's artifacts, samples, or outputs; separate confirmation-only B

ID / priority	Gap	Required acceptance before relying on the result
A1 / P0	Audit gates enforce cutoff one instead of two	One declared cutoff applied to all three partition pairs; reject distance 0/1/2 fixtures and accept an isolated distance-3 case
M1 / P0	Undefined null passes; low-coverage correlation is replaced by zero	Require finite statistics and adequate support; report undefined values, reasons, and coverage rather than fabricated zeros
D2 / P1	Order-dependent capped pairs, null policy, unconditional conclusion	Filter unique eligible pairs before capping; specify seeded ordering, self-match policy, support, uncertainty, and a conclusion rule or descriptive-only reporting
V1 / P1	Incomplete schema/assignment guarantees	Enforce exact 9-mer/alphabet/label contracts, unique and complete keys, allowed split names, no extra assignments, valid allele mapping, and proper secondary scope
P1 / P1	Mutable outputs, import-time execution, incomplete run provenance	Separate loading/checking/generation; use isolated fixtures and pure helper seams in future tests; record versions, hashes, seeds, command, and status

These findings do **not** demonstrate that the saved split contains a distance-two violation. The stored report records zero evaluation-to-training neighbors at the configured cutoff; upstream prose reports a closest cross-partition distance of three.[46][49] Keep those as recorded findings, not a fresh exhaustive audit. Do not reroll the split to repair the narrative.

3. Selected architecture: GeoStab-FT

This is a project-specific engineering design, not a claim of first-in-literature novelty.



Why this genuinely adapts a foundation model

LoRA trains low-rank matrices inside ESM-2:

$$W_{\text{effective}} = W_{\text{pretrained}} + (\alpha / r) B A$$

Original matrices stay frozen, but the effective encoder and its representations change under the half-life loss. This is parameter-efficient fine-tuning, not merely a head on old embeddings.

The partial-unfreezing control updates original weights in selected blocks. It tests whether additional adaptation cost is justified. Boltz stays a fixed coordinate generator; scalar half-life labels are not coordinate supervision for an end-to-end structure-model fine-tune.

Serova-listed model choices

Role	Selection	Boundary
Primary sequence foundation	facebook/esm2_t30_150M_UR50D	Moderate-size adaptation target with verified implementation
Smaller fallback	facebook/esm2_t12_35M_UR50D	Only after a measured resource failure; run matched frozen/adapted controls
Primary 3D foundation	Boltz-2	Coordinates and confidence only; no peptide-as-small-molecule affinity
Optional inverse-folding feature	ProteinMPNN	Peptide-only score on validated backbones, not a half-life output
Permitted but deferred alternatives	ESMC, ProtT5, SaProt, Chai-1, Protenix, ESMFold/ESMFold2, LigandMPNN, ESM-IF	Avoid a tokenizer/model/structure-protocol sweep on the critical path

Do not use target-trained MINT/SPEARMINT or NetMHCstabpan weights as independent baselines. Supporting papers do not replace the challenge's model/data choice.

Verified pins and dimensions

- ESM-2 150M revision: a695f6045e2e32885fa60af20c13cb35398ce30c.
- ESM-2 35M fallback revision: 6fbf070e65b0b7291e7bbcd451118c216cff79d8.
- 150M configuration: 30 layers, width 640, 20 attention heads, intermediate width 2560, maximum-position configuration 1026.
- Inspected Transformers ESM source: v4.57.1, blob 63d9344188cc83bd8ac4719db78def849afbda7f. Projections are query, key, value; blocks are encoder . layer.
- Inspected Boltz source: b1ebfc46ecf57f5414e0d1a6f9027bbb122c53bc; prediction-doc blob 7ec361693074e76042ccb4a30503c3f1e99e387a.

Pin the compatible environment and exact Boltz weight checksum during implementation preflight. These pins are not a claim that an uninstalled stack has passed a GPU smoke test.

4. Dataset and evaluation contract

Use **only** context/dataset.csv **for core supervised training**: allele, peptide, thalf_hours, hla_seq, hla_pseudoseq. No extra DTU half-life download, affinity corpus, synthetic targets, or published predictor outputs are required.

The supplied HLA sequence is a 182-aa domain, not a mature full construct. Preserve engineered suffixes and distinguish pseudosequence collisions.

Reuse the seed-42 Hamming≤2 component split and its cluster_id:

Partition	Rows	Distinct peptides	Alleles
Train	22,532	4,514	75
Validation	2,817	563	73
Test	2,817	556	75

The stored report records 5,410 components. These are retained saved facts, not a fresh audit. Do not resplit to match the proposal's approximate paper counts.

Require exact schemas, valid nine-residue peptides, finite nonnegative labels, unique keys, complete many-to-one joins, and no distance-0/1/2 crossings across any partition pair.

Existing gaps remain:

- The local audit's pass/fail distance checks still enforce only one-edit separation.
- The reviewed newer upstream diagnostic uses whole-dataset normalization before its nominal training-only analysis.
- Undefined null correlations and zero sentinels require explicit handling.
- Historical evaluation-label-informed split design remains disclosed.

The future training pipeline must use its own training-only preprocessing, never the diagnostic's global z table. Harden checks before scientific claims without regenerating the benchmark.

The existing allele split is secondary: it holds out alleles, not necessarily peptides, and uses label-informed stratification. It is not external-assay validation.

5. Exact sequence-model specification

Encoder and interaction head

Encode peptide and HLA separately with the **same** ESM instance. Do not invent chain tokens or concatenate them as a fictitious biological chain.

Remove BOS/EOS/padding positions:

```
E_p: [B,9,640]
E_h: [B,182,640]
P = Linear_p(E_p) + learned_peptide_position[0:9] # width 128
H = Linear_h(E_h)                               # width 128
```

- One peptide-query/HLA-key-value cross-attention block.
- Width 128; four heads of 32 dimensions.
- Residual connection, LayerNorm, dropout 0.1.
- Flatten nine output tokens and concatenate masked mean projected HLA: $9 \times 128 + 128 = 1280$ features.
- Head: $\text{Linear}(1280, 128) \rightarrow \text{GELU} \rightarrow \text{Dropout}(0.1) \rightarrow \text{Linear}(128, 1)$.
- Scalar output μ_{seq} represents $\log_{10}(\text{hours})$.
- Log actual head parameter counts; attention is not automatically causal explanation.

Adaptation arms

ID	Backbone policy	Head
F150	Frozen 150M	Pooled-sequence MLP
X150	Frozen 150M	Residue interaction head above
L150 — primary	LoRA rank 8, alpha 16, dropout 0.05, bias none, K/V projections in all 30 blocks	Same interaction head
U150 — budgeted comparator	Unfreeze original blocks 28/29 and <code>encoder.emb_layer_norm_after</code> ; freeze other originals; no LoRA	Same interaction head

Attach LoRA only to the backbone before adding the head. Assert 60 targeted projection modules. Expected adapter-matrix count:

```
30 × 2 × 8 × (640 + 640) = 614,400 trainable weights
```

This calculated count excludes the head; assert it against the real model. Enumerate U150's actual trainable names/counts rather than claiming measured memory now.

For a 35M fallback, derive hidden width, layer count, projection dimensions, and adapter-count assertions from that pinned configuration. Do not reuse 150M-specific dimensions, the 60-module assertion, or its embedding caches.

Do not presume LoRA always wins or is always faster. If U150 cannot fit the approved budget, report it as unrun; L150 still provides genuine adaptation.

Cache contract

- F150/X150 may reuse versioned frozen residue embeddings.
- L150/U150 must recompute embeddings affected by each optimizer update. A detached across-step HLA cache invalidates fine-tuning.
- Within-minibatch deduplication is allowed only if gradients remain correct.
- After freezing the selected adapted model, rebuild its cache for structural training/inference and key it by adapter/model checksum.
- Never substitute old frozen-base embeddings for the adapted encoder.
- Validation/test prediction uses evaluation mode and no gradients.

6. Training recipe and objectives

```
y = log1p(thalf_hours)          # after validation
loss = mean((mu_seq - y)^2)
hours_point = max(expm1(mu), 0) # declared operational convention
```

Report log errors before presentation clipping and operational hours errors after the same rule for all models. $\text{expm1}(E[\log1p(t)])$ is not generally $E[t]$; do not call it an unbiased expected half-life in hours.

Do not add a censored/zero head by default: the CSV does not establish a detection limit. A hurdle model is a separate empirical experiment, not automatic censoring correction.

Setting	Proposed initial value
Optimizer	AdamW, trainable parameters only
LoRA / head learning rate	$2e-4$ / $1e-3$
U150 original-weight learning rate	$1e-5$; same head rate
Weight decay	$1e-2$; exclude biases/normalization
Microbatch / accumulation	8 pairs / 4 steps; effective batch 32
Precision	CUDA bfloat16 where supported; record actual setting
Checkpointing	Non-reentrant if needed; verify adapter gradients
Schedule	10% optimizer-step warmup, then cosine decay
Gradient clipping	Norm 1.0
Maximum epochs	8 plus a separate runtime/spend cap
Validation/checkpoint selection	Once per epoch; finite validation Spearman, log-RMSE tie-break

Setting	Proposed initial value
Early stopping	Patience 2 validation evaluations
Seeds	Primary 42; predeclared replication 43/44 only if budget permits

These are fixed starting decisions, not an unbounded sweep. Record OOM/numerical changes and keep relevant comparison policies aligned. Undefined validation correlation stays undefined; log-error may be used for checkpoint bookkeeping without inventing correlation.

Required controls: B0 training-only global/allele median; B1 small supervised positional sequence model with allele identity; F150 pooled frozen; X150 frozen interaction; L150 adapted interaction. Add a matched one-hot-allele replacement where feasible.

B1's default is a positional one-hot MLP: 180 peptide values, 680 pseudosequence values, and a training-fitted allele indicator vector (75 values on the primary split) feed `Linear(935,128) → ReLU → Dropout(0.1) → Linear(128,32) → ReLU → Dropout(0.1) → Linear(32,1)`. Derive the input width from the training vocabulary for other analyses; unseen alleles use an all-zero identity vector, not a new untrained category weight. Use the same log-MSE target, head AdamW settings, and validation selection rule, with at most 30 epochs and patience 5 for this cheap baseline; report its distinct training budget.

F150 concatenates masked mean peptide and HLA embeddings into 1280 values and uses `Linear(1280,128) → GELU → Dropout(0.1) → Linear(128,1)`. X150/L150 retain the residue-interaction head and matched eight-epoch maximum specified above.

A frozen 35M result alone is not a fair comparator for adapted 150M. A 35M fallback requires its own matched frozen/adapted arms.

7. Boltz-2 structures and bounded S0 cohort

Verified construct protocol

- Resolve names/engineered status against a pinned official HLA source.
- Require exact identity over the sponsor's domain and a recorded mature extracellular heavy chain plus mature β 2m and peptide as separate protein chains.
- Preserve the prior domain-mismatch and C67S quarantines until resolved. Do not substitute a supertype or precursor silently.
- PDB references support mapping/QC/exposure, not half-life labels. The Serova figure cites an eight-residue epitope; do not assume it is a matching nine-mer template.

The previously reported domain discrepancies are HLA-B*08:03, HLA-A*02:50, and HLA-A*24:19. The separate engineered group is HLA-B*14:01(C67S), HLA-B*14:02(C67S), and HLA-B*39:06(C67S). Preserve valid sequence-experiment rows; unresolved structural constructs remain ineligible until reconciled against the pinned source.

Primary inference protocol

Propose one pose per pair, three recycling steps, 200 sampling steps, one diffusion sample, `max_parallel_samples=1`, mmCIF output, and **no affinity property**. These are documented options/defaults, not a runtime promise.

Pin code and weights. Cache heavy-chain/ β 2m MSAs by exact construct/source; the peptide may use explicitly declared `msa: empty`. If using the documented MSA service, record public-sequence upload and output provenance. Single-sequence mode is not silently equivalent to MSA-assisted prediction.

Verify chain/token mapping and confidence scales; do not assume peptide chain C. Official confidence JSON scores are described on [0,1]; arbitrary experimental B-factors are not pLDDT.

A fixed $K=3$ ensemble is a later ablation, not core throughput. Spread is model/template sensitivity, not measured physical entropy or an off-rate.

Candidate membership inventory

Only metadata, not half-life labels, was used:

Allele	Train rows/components	Validation rows/components	Test rows/components
HLA-A*02:01	821 / 786	102 / 99	100 / 100
HLA-B*15:01	856 / 834	120 / 113	94 / 90

These are proposed pilot alleles subject to construct verification, not evidence of all-allele structural coverage.

After a separately approved three-pair training-only cost probe:

- Target at most **512 train**, **64 validation**, **128 test rows** within the existing partitions.
- Select whole peptide components among eligible candidate-allele rows, seed 42, label-free and approximately allele-balanced.
- Counts are caps, not promised exact counts. Retain whole components and record actual support; freeze IDs before inference.
- If support or budget is too small, label the structural result a pilot/unrun. Never replace failed test cases with easier ones.
- Every control shares the cohort, attempt policy, and available confidence information.

Structural validity

Freeze label-blind QC after training-only probes/reference inspection. The proposed initial hard checks require correct sequences/maps, finite coordinates, N/CA/C/O at every peptide position, and a finite mapped C α for every one of the 182 HLA-domain positions. Preserve heavy-atom completeness masks separately.

Initial engineering pose checks: both peptide termini must have a mapped HLA heavy atom within 8 Å; at least seven of the nine peptide positions must have one within 8 Å; and no peptide–HLA heavy-atom pair may be closer than 1 Å. These are coarse invalid-pose checks, not proof of a correct binding register. Inspect them on training-only probes/reference examples and version any justified revision before cohort inference. Never change them using held-out prediction errors.

Low confidence is separate from hard invalidity and enters the confidence control. Missing/invalid structure always has a reason and exact sequence-model fallback.

8. Geometry-aware head and chemistry

Mapped geometry

```
D[i,j] = minimum heavy-atom distance from peptide residue i
         to mapped HLA-domain residue j
D, valid_pair_mask: [9,182]
```

Use only peptide–mapped-HLA contacts, not β 2m or intrachain contacts. Indices must align with the sequence embeddings.

Six Gaussian radial features use centers [3,4,5,6,8,10] Å, width 1.5 Å, and zero features beyond 12 Å. Missing pairs contribute zero geometric bias with explicit masks. These are engineering definitions, not universal binding constants.

Precisely, $\text{RBF}_k(d) = \exp(-0.5 * ((d - \text{center}_k) / 1.5)^2)$ for valid distances $0 \leq d \leq 12$ Å, and zero otherwise. Set invalid/missing entries to a safe value before arithmetic, then mask their features to zero so NaNs cannot enter attention.

Learned attention bias

Reuse the frozen selected sequence model and attention projections:

```
G_h[i,j] = 2 * tanh(sum_k w[h,k] * RBF_k(D[i,j]))
A_h = softmax(Q_h K_h^T / sqrt(32) + G_h + padding_mask)
```

Train w : [4,6], initialized to zero. Geometry therefore changes peptide–HLA attention rather than only decorating the demo. No gradient enters Boltz.

Freeze the full sequence model before this stage, preventing a small structural cohort from overwriting the full-data adapter and making fallback exact.

Choose the common structural-stage base S^* from the completed X150/L150/U150 interaction-head arms using the declared validation rule, then freeze it before structural-head fitting. All three expose compatible residue attention. F150 and B0/B1 remain comparators, not interchangeable geometry backbones. Record $S^*_{\text{model_id}}$; if X150 wins, report that the selected base is frozen rather than calling it adapted.

Matched residual comparison

- Confidence vector: mapped peptide–HLA interface confidence, mean peptide pLDDT, mean mapped-domain pLDDT; verify output maps/scales.
- $z = \text{concat}(\text{flatten}(\text{nine attention outputs}), \text{mean}(\text{projected HLA})), \text{width } 1280$.
- **G_conf**: base sequence prediction plus a regularized linear residual on normalized $[z_{\text{without_geometry}}, \text{confidence}]$.
- **G_pair**: identical residual architecture, initialization, valid training rows, and selection budget, using $z_{\text{with_geometry}}$ plus the 24 geometric-bias parameters.
- Use `LayerNorm(1283, elementwise_affine=False)` on each example's 1280 context values plus three confidence values, followed by the linear residual. No all-label normalization is involved.

- Initialize residual output to zero; verify gradients after more than one step because the initial zero residual can temporarily suppress upstream gradients.
- Proposed residual recipe: AdamW $1e-3$, weight decay $1e-2$, batch 64, at most 200 epochs, validation patience 15, and runtime cap.
- Both arms return exactly the same frozen sequence prediction for invalid/missing test structures.

Train availability/QC-only diagnostics on all S0 training rows separately. Geometry/confidence comparisons use identical valid training eligibility. Report full-S0 operational and matched-valid-subset results, with selected/attempted/valid/fallback counts.

Fixed-contact control

Retain the proposal's useful 9×20 feature idea:

```
C[i,a] = sum_{j: HLA_type[j]=a} sigmoid(2*(5-D[i,j])) / N_mapped
```

Save raw sums, denominator, masks, and all nine positions. Fit/evaluate its residual control on the same eligible cohort/fallback policy. Compare this cheap fixed representation with learned geometric attention.

This tensor does not guarantee unique allele identity or eliminate every possible collision. Preserve full sequence/provenance regardless.

Hydrogen-bond and terminal-geometry ablation

- Do not infer verified hydrogen bonds from only a 2.8–3.2 Å heavy-atom distance.
- Biotite's documented Baker–Hubbard method requires explicit hydrogens, H–acceptor distance ≤ 2.5 Å, and donor–H–acceptor angle $>120^\circ$ by default.
- Hydrogen placement, terminal states, protonation, and chemically valid donor/acceptor selection require an explicit validated protocol.
- Without those prerequisites, report the block unavailable or call a specified heavy-atom descriptor “polar-contact geometry,” not a hydrogen-bond count.
- Map mature-chain numbering before naming HLA clamp residues; do not blindly transfer the proposal's residue-number list.
- Test terminal interactions/burial while retaining all positions and allele-specific anchor exceptions.
- SASA and pose dispersion are descriptors, not direct entropy/kinetic measurements.

Optional ProteinMPNN features retain peptide-only masks, fixed receptor context, score normalization, and a fixed-template sensitivity control. They remain compatibility hypotheses, not half-life labels.

9. Hypotheses and acceptance of scientific results

Hypothesis	Comparison	What can be concluded
H1 — adaptation helps	L150 vs X150 on identical full-dataset test pairs	Incremental LoRA utility under this protocol
H2 — interaction head helps	X150 vs F150, with capacity/tuning/cost disclosed	Utility of the residue-level design, not proof of mechanism

Hypothesis	Comparison	What can be concluded
H3 — original-weight updates justify cost	U150 vs L150/X150	Accuracy–cost evidence; explicitly unrun if unaffordable
H4 — 3D adds beyond confidence	G_pair vs G_conf with identical S0/fallbacks	Incremental geometry signal, not static-pose proof of kinetics
H5 — learned geometry beats fixed contacts	G_pair vs contact-tensor residual	Architecture comparison with eligibility/capacity disclosed
H6 — chemistry/ensemble/inverse features help	Separate optional add-one-block comparisons	Endpoint-specific evidence, not entropy/censoring/clinical guarantees

Primary held-out measure: Spearman with paired peptide-component bootstrap differences. Also report log-MAE/RMSE, operational hours-MAE, supported within-allele metrics, counts, seed variation where run, GPU cost, latency, coverage, and failures.

Spearman does not establish accurate absolute hours; Pearson alone does not establish calibration. A high score is not by itself proof of leakage or correct evaluation. Published SPEARMINT numbers are not expected values for this model/split.

Fit/select using train/validation only; freeze every comparison before final test reporting and retain negative results. An interval spanning no gain is inconclusive, not equivalence. A structural pilot is not full-corpus evidence.

Resample components with paired predictions, not independent samples or rows. Separate test-sampling uncertainty from retraining variation.

For the planned primary interval calculation, use 2,000 component-bootstrap draws with seed 2026 and percentile bounds at 2.5%/97.5%. Keep all rows belonging to each drawn component and both models' aligned predictions. Record undefined draws rather than silently replacing/resampling them; report the interval unavailable if fewer than 95% of draws are valid. This is a prespecified reporting guard, not a universal inference theorem. Report log-RMSE and hours errors alongside rank changes, and name the endpoint actually improved.

Do not show per-prediction intervals without a valid separate calibration design. Reusing repeatedly tuned validation labels for calibration does not automatically provide coverage; metric confidence intervals are not individual prediction intervals.

10. Modal feasibility and spending

Starting hardware proposals

- Sequence: one **A10 24 GB** worker (gpu="A10" in the inspected Modal API), microbatch 8/effective batch 32.
- On OOM, halve microbatch and increase accumulation to preserve effective batch, down to one before declaring failure.
- A larger GPU or 35M fallback requires an explicit recorded decision within the approved budget and matched controls.
- Structures: one **L40S 48 GB** worker (gpu="L40S") for the probe; measure pHLA workload rather than copying ligand benchmark throughput.

- Separate sequence and Boltz container environments; versioned Modal Volume artifacts with explicit commits.
- Initially one GPU per workstream; at most two concurrent only after cap approval. No assumed 256-way fan-out.

These are profiling targets, not verified memory-fit or completion-time guarantees.

Published rates inspected

Resource	Rate
A10	\$0.000306 / second
L40S	\$0.000542 / second
A100 40 GB	\$0.000583 / second
A100 80 GB	\$0.000694 / second
Physical CPU core	\$0.0000131 / core / second
Memory	\$0.00000222 / GiB / second

Recheck before launch. These are public base rates, not the user's balance, final quote, or authorization; CPU, memory, storage, region, retries, and account settings matter.

The separate memory meter refers to host RAM; do not charge GPU VRAM again on top of the GPU rate when constructing the estimate.

No numerical cap was confirmed. Define `B_approved` only after authorization and usable-credit verification. Proposed allocation: 10% preflight, 40% sequence comparisons, 30% structures/residuals, 10% final evidence/demo, 10% reserve.

```
cost = GPU_seconds*GPU_rate
      + CPU_core_seconds*CPU_rate
      + memory_GiB_seconds*memory_rate
      + known storage/other charges
```

The later probe measures VRAM, throughput, cold-start/download overhead, checkpoint cost, and retries. Derive epochs/cohort size from those measurements, not nominal credit offers.

Persist reservations before launch; cap job time, attempts, concurrency, and total admitted cost. Do not rely only on delayed dashboard alerts. Checkpoint/resume interrupted work, use idempotent pair keys, and do not duplicate an uncertain in-flight paid job.

11. Future implementation map — not files to create now

Proposed file	Responsibility / interface
src/data.py	load_frozen_dataset(csv_path, split_path): schemas/hashes/keys/labels/partitions, preserving component IDs

Proposed file	Responsibility / interface
src/models/encoder.py	Shared pinned ESM, frozen/LoRA/last-two policies, parameter ledger, residue masks, adapter save/load
src/models/interaction.py	forward(E_p,E_h,masks,distance_features=None): 128-wide four-head sequence/geometric interactions
src/models/stability.py	Sequence and matched residual predictors, log/hour outputs, provenance and exact fallback
src/train.py	Bounded accumulation/checkpointing, finite gradients, validation-only selection, resume
src/structure_inputs.py	Verified constructs/chains, domain maps, immutable pair IDs, Boltz YAML/MSA provenance
src/fold.py	Pinned bounded inference, idempotent outputs, attempts/failures/runtime/VRAM/cost
src/interface_features.py	Mapped distances/RBFs/contact tensor/masks, confidence, optional validated chemistry
src/evaluate.py	Pair equality, full/valid cohorts, undefined metrics, component bootstrap and allele support
src/inverse_score.py	Optional ProteinMPNN peptide scores/provenance
src/app.py	Real-output demo with correct model/pose/fallback labels and no invented intervals
modal_app.py	Separate images/caches, profiling, bounded jobs and explicit budget admission
configs/experiment.yaml	Pins/hashes/trainable scope/seeds/recipe/cohort/features/cost/time caps
pyproject.toml and lockfile	Minimal compatible tested dependencies, no floating latest installation

Batch contracts: peptide [B,9], HLA [B,182], masks, pair/component IDs, labels only for allowed partitions, optional RBFs [B,9,182,6], and explicit geometry availability.

Cache keys include sequence/construct hash, base/adaptor revision, feature/protocol version, and seed. Do not commit raw model weights or coordinate dumps.

A later code-hardening task also touches scripts/audit_splits.py and deliberately reconciles scripts/measure_leakage_by_distance.py for threshold, isolation, sampling, and statistic guards. Do not run the generator on frozen outputs; any generator maintenance preserves existing assignments. None of these code edits is included in this documentation approval.

12. Future verification

Area	Required tests
Data/splits	Exact keys/schema; invalid labels/alphabet/length; all partition pairs reject distances 0/1/2; distance-3 fixture; training-only preprocessing

Area	Required tests
Tokenization	BOS/EOS/padding exclusion, lengths, masks, batch invariance, sequence order
LoRA scope	60 K/V modules and 614,400 adapter weights; originals unchanged; adapter gradients and updates after two steps
Partial scope	Only blocks 28/29, final norm, and head update
Caches	Adaptation invalidates old embeddings; frozen cache matches live inference
Interaction	Shapes/masks; zero geometry bias equals sequence block; no fictitious chain semantics
Geometry	Exact synthetic distances/RBFs, rigid invariance, index maps, masks, no β_2 m/intrachain contamination
Chemistry	Explicit-H positive/negative and angular boundary fixtures; missing hydrogens means unavailable, not zero bonding
Residuals	Matched training eligibility/pair order, frozen common sequence model, zero-initial identity, exact invalid-structure fallback
Metrics	Known toy errors/ranks, undefined cases, component resampling, log/hour semantics, denominators
Reproducibility	Tiny synthetic ESM CPU overfit/save-load tests; later approved GPU finite-gradient/resume smoke test
Budget	Dry-run admission, timeouts/retries, no job without cap, no duplicate uncertain jobs

Use tiny synthetic model configurations for CPU tests rather than downloading real weights by default. Real checkpoint/GPU tests are separately gated. No model build/test suite runs for the current document update.

13. Delivery gates

Gate	Output	Stop/fallback
D0 — final local master plan	Architecture, Serova alignment, sources, controls, recipe, boundaries	No implementation/spending implied
C0 — integrity	CPU tests, immutable split loading, trainable/cache/metric checks	Correctness before scores
G0 — approved probe	Cap/access/images, measured VRAM/throughput/cost, tiny-batch fit	Reduce microbatch, explicit matched smaller model, or stop
G1 — adaptation	B0/B1/F150/X150/L150, budgeted U150, validation-selected checkpoints	Report negative adaptation; no guaranteed baseline score
G2 — structural feasibility	Exact constructs, three training-only probes, mapping/QC/features	Explicit no-go rather than silent supertype substitution
G3 — structural experiment	Frozen S0, all attempts, matched G_conf/G_pair; contact-tensor control if run	Pilot/unrun if unsupported; retain sequence fallback and report incomplete 3D work as such

Gate	Output	Stop/fallback
G4 — evidence	Frozen test predictions, paired effects, support/failures/cost	No post-test model selection
G5 — demo	Real point predictions and labeled predicted/reference poses	Cached real outputs/recording, no fabricated coverage/dynamics

The official context is October 3–4. The earlier recorded October 4, 14:45 BST deadline remains an organizer-rule reference needing confirmation; the new extraction did not verify it. The proposal's expired 16:30/20:00 checkpoints and unmeasured runtime guarantees are not operational commitments. Execution follows the remaining resources and gates rather than a promised completion time.

14. Proposal reconciliation

Build-proposal claim/idea	Final treatment
Clear question, paired ablations, sequence fallback	Keep
Residue-level contacts and terminal chemistry	Keep with validated atom/mapping/normalization contracts
“Nobody fine-tunes”	Supersede: user requests adaptation and Serova permits it
Embeddings computed once remain reusable	Only while encoder/adapters stay unchanged
Structural spread equals entropy/kinetics	Optional predictive hypothesis, not a physical measurement
Guaranteed baseline or borrowed expected rho	Remove; no executed result on this configuration/split
Match paper counts for caveat-free comparison	Remove; retain Serova CSV and actual frozen split
NetMHCstabpan as an honest held-out baseline	Exclude, per Serova
Zero proves censoring; inverse-log hurdle output is expected hours	Remove unsupported observation/expectation claims
Automatic cheap affinity-transfer gain	Defer external labels and task-specific transfer/decontamination
Canonical backbone or supertype guarantees correctness	Require exact construct and structure validation
28k×3 structures with 256-way fan-out	Replace with measured, budget-capped S0
Heavy-atom distance proves H-bond; attribution proves mechanism	Correct chemistry and interpretation
Mandatory conformal intervals from reused validation	Separate valid calibration design or no interval
Any rho above 0.85 necessarily proves leakage	Systematic auditing, not a universal score threshold

15. Current architecture sources

Sources A1–A19 support the current design. The retained script-audit and historical findings use their original bracketed reference numbers in §17; those are a separate reference set, not new experiment results.

- **A1: Authoritative Serova challenge:** [repository PDF](#), byte-identical to the user-supplied Serova Protein Engineering Track Challenge.pdf; §§3–6 define models, data, fine-tuning options, and evaluation.

- **A2: Official event:** <https://iterate.inc/london-ai-science?welcome=true> — extracted successfully; Resources extraction added no detailed rules.
- **A3: User-supplied proposal:** Track 3 Build Plan – Structure-Aware pHLA Stability Prediction.pdf, 25 pages, SHA-256 a0e212c91ec8565c86e436a63b6b9269f6a2961d2c6161541885fae3f6cab0bc. Read as a design proposal; it is not copied into this repository or treated as proof of performance.
- **A4: Dataset:** Rasmussen et al. (2016), <https://doi.org/10.4049/jimmunol.1600582>
- **A5: Pinned ESM-2 150M:** https://huggingface.co/facebook/esm2_t30_150M_UR50D/tree/a695f6045e2e32885fa60af20c13cb35398ce30c
- **A6: ESM source:** <https://github.com/facebookresearch/esm>
- **A7: Inspected ESM implementation:** https://github.com/huggingface/transformers/blob/v4.57.1/src/transformers/models/esm/modeling_esm.py
- **A8: Peer-reviewed fine-tuning evidence:** <https://www.nature.com/articles/s41467-024-51844-2> — different tasks, not Serova results.
- **A9: PPI PEFT/K-V study:** https://www.mlsb.io/papers_2023/Parameter-Efficient_Fine-Tuning_of_Protein_Language_Models_Improves_Prediction_of_Protein-Protein_Interactions.pdf — workshop classification evidence, not a guaranteed regression gain.
- **A10: PEFT guidance:** https://huggingface.co/docs/peft/main/en/developer_guides/lora
- **A11: Supplementary ESM-LoRA example:** <https://huggingface.co/blog/AmelieSchreiber/esmbind> — worked example with similarity caveats.
- **A12: Pinned Boltz protocol:** <https://github.com/jwohlwend/boltz/blob/b1ebfc46ecf57f5414e0d1a6f9027bbb122c53bc/docs/prediction.md>
- **A13: ProteinMPNN:** <https://github.com/dauparas/ProteinMPNN>
- **A14: Related structural classification:** <https://www.frontiersin.org/journals/immunology/articles/10.3389/fimmu.2020.01583/full> — not equivalent quantitative half-life evidence.
- **A15: Hydrogen-bond requirements:** <https://www.biotite-python.org/latest/apidoc/biotite.structure.hbond.html>
- **A16: Modal public rates:** <https://modal.com/pricing>
- **A17: Official HLA provenance:** <https://www.ebi.ac.uk/ipd/imgt/hla> and <https://github.com/ANHIG/IMGTHLA>
- **A18: Tavily deep-research request:** 95a53a2a-75d8-4631-825e-e4c5fa2ac6cc; discovery/synthesis only, with consequential claims cross-checked.
- **A19: Modal GPU identifiers and hardware guidance:** <https://modal.com/docs/guide/gpu> — current documented A10/L40S identifiers and L40S memory, not a workload-specific fit guarantee.

16. Retained artifact and historical research evidence

Input identity

Artifact	SHA-256 recorded for this revision
context/dataset.csv	384e0accdd35c589f02bd3f7d702d31b00a27f8f6f728cacb52b4d8a6c83af44a

Artifact	SHA-256 recorded for this revision
splits/peptide_split.csv	0d92a33bb64eb35d922f698283ca08f6556019e55c7f87c7ee5d31ab6b4d7988
splits/allele_split.csv	6632491f72c3af890b82c4e5753863866089ae43cc0dd3b5d0af1c5bd521bdd1
splits/split_report.json	b9893123ef40bacc28367b20e2eabec7bb8b7d9bd1bb3d0a41d87d160e28b539

Their Git blobs match the inspected upstream snapshot. Matching counts between the earlier exported-sheet audit and the current CSV do not establish byte identity of an unavailable earlier export. Bind future experiments to these current input identities; retain earlier findings with attribution.

Prior research findings retained with their boundaries

These findings were reported by the earlier plan and its cited sources. They were **not recomputed during this script/documentation review**:

- The inspected SPEARMINT consensus reference maps 72/75 sponsor allele names; the missing names are the three C67S constructs. Separately, HLA-B*08:03, HLA-A*02:50, and HLA-A*24:19 have domain discrepancies against that reference. Those two categories are not the same three alleles. Reconcile against a pinned official sequence source rather than assuming the sponsor input is wrong.[6][42]
- Earlier pilot mapping found full-chain lengths 365 for HLA-A*02:01 and 362 for HLA-B*15:01, with exact supplied-domain matches starting at zero-based offset 24. These are prior reference observations, not permission to use a precursor as the mature structural construct.[6][42]
- SPEARMINT describes 27,034 curated measurements across 72 alleles, unlike the sponsor's 28,166 rows and 75 alleles. Its released train/validation/test counts are 21,626/2,705/2,700, totaling 27,031; the three-row discrepancy remains a limitation on claims of exact raw-to-split reproduction.[1][4]
- The preprint reports MINT transfer Spearman 0.791 versus ESM-2 transfer 0.745. Earlier reanalysis of 2,700 released Stage-2 test predictions reported MINT 0.7906 and NetMHCstabpan 0.8763; the latter is not a clean held-out comparator. Hours-scale CCC was 0.6735/0.4997, versus log-hours CCC 0.7879/0.7860. Always state the scale.[1][4]
- The earlier overlap audit reported all 2,700 Stage-2 test pairs in the sponsor CSV and 568 also in Stage-3 training. Of 1,133 Stage-3 test records, 979 had a reference identifier seen in training, leaving 154 new-reference rows. Peptide-disjoint testing is not automatically new-study testing.[1][4]
- Earlier reanalysis of released MINT_stage3_film_v2 predictions reported SPA Spearman 0.3591 (217 rows), purified fluorescence 0.5666 (758), and cellular fluorescence 0.2204 (158). These are prediction-file reanalyses, not fresh inference; other published variants are different methods. MINT's interaction pretraining and architecture also prevent attributing its difference from ESM-2 solely to cross-chain attention.[1][4][22]

The referenced SPEARMINT code/prediction snapshot is d548dee9b04dd0e3a9d7a50a593d2c030e31d2a5. An attributed reproduction can be useful, but released stability-trained checkpoints are not the primary blind baselines on this sheet. A retrospective uncertainty analysis on those predictions would be a separate research task, not a replacement for B1/F0 here.[1][4][7]

17. Prior sources referenced by the retained evidence

[1] <https://www.biorxiv.org/content/10.64898/2026.06.28.735023v1.full.pdf> [2] <https://iterate.inc/london-ai-science?welcome=true&tab=home> [3] <https://iterate.inc/london-ai-science?welcome=true&tab=resources> [4] <https://github.com/pirl-unc/spearmint> [5] <https://drive.google.com/file/d/11i-LSRZlgLS7T2IVM-12xuddwuryDmnu/view?usp=sharing> [6] <https://docs.google.com/spreadsheets/d/1NiZNvcF3u0KF-1bbuA50CF3IXvs1l4HfbaR3KRLvso/edit?usp=sharing> [7] <https://huggingface.co/dkarthikeyan1/spearmint> [8] <https://arxiv.org/abs/1905.03222> [9] <https://www.nature.com/articles/s41467-025-67971-3> [10] <https://pmc.ncbi.nlm.nih.gov/articles/PMC4976001> [11] <https://modal.com/docs/guide/batch-processing> [12] <https://modal.com/docs/guide/gpu> [13] https://huggingface.co/facebook/esm2_t12_35M_UR50D [14] https://huggingface.co/facebook/esm2_t30_150M_UR50D [15] <https://amass.tech> [16] <https://antigravity.google/use-cases/science> [17] <https://arxiv.org/abs/1904.06019> [18] <https://docs.google.com/presentation/d/1zpERXrvfL2akHcEySw7gJxSWedW-3K3uk2kobw4jtDY/edit?usp=sharing> [19] <https://luma.com/3iipivod> [20] <https://www.biorxiv.org/content/biorxiv/early/2026/06/29/2026.06.28.735023/DC1/embed/media-1.pdf?download=true> [21] <https://doi.org/10.1016/j.immuno.2023.100030> [22] <https://github.com/VarunUllanat/mint> [23] <https://github.com/jwohlwend/boltz/blob/main/docs/prediction.md> [24] https://github.com/aswin-giridhar/london-ai-science-hackathon/blob/main/context/serova_primer.pdf [25] <https://github.com/dauparas/ProteinMPNN> [26] <https://github.com/dauparas/LigandMPNN> [27] <https://www.frontiersin.org/journals/immunology/articles/10.3389/fimmu.2020.01583/full> [28] <https://www.frontiersin.org/journals/immunology/articles/10.3389/fimmu.2022.878762/full> [29] <https://pmc.ncbi.nlm.nih.gov/articles/PMC13106973> [30] <https://github.com/X-lab-3D/swiftmhc-inference> [31] <https://www.sciencedirect.com/science/article/pii/S0969212623004136> [32] <https://github.com/facebookresearch/esm> [33] <https://zenodo.org/records/15495898> [34] <https://github.com/chaidiscovery/chai-lab> [35] <https://github.com/bytedance/Protenix> [36] <https://github.com/westlake-repl/SaProt> [37] <https://github.com/X-lab-3D/PANDORA> [38] <https://github.com/Biohub/esm> [39] <https://github.com/agemagician/ProtTrans> [40] <https://github.com/aswin-giridhar/london-ai-science-hackathon/blob/main/context/NOTES.md> [41] <https://www.kavrakilab.org/publications/abella2020-pnas.pdf> [42] https://raw.githubusercontent.com/pirl-unc/spearmint/d548dee9b04dd0e3a9d7a50a593d2c030e31d2a5/refs/2field_hla_consensus_seqs.csv [43] https://github.com/aswin-giridhar/london-ai-science-hackathon/blob/de2532f0a0d97067828dfd0077d0b852a845b920/scripts/make_splits.py [44] https://github.com/aswin-giridhar/london-ai-science-hackathon/blob/de2532f0a0d97067828dfd0077d0b852a845b920/scripts/audit_splits.py [45] https://github.com/aswin-giridhar/london-ai-science-hackathon/blob/de2532f0a0d97067828dfd0077d0b852a845b920/scripts/measure_leakage_by_distance.py [46] https://github.com/aswin-giridhar/london-ai-science-hackathon/blob/de2532f0a0d97067828dfd0077d0b852a845b920/splits/split_report.json [47] <https://github.com/aswin-giridhar/london-ai-science-hackathon/blob/de2532f0a0d97067828dfd0077d0b852a845b920/RECONCILIATION.md> [48] <https://github.com/aswin-giridhar/london-ai-science-hackathon/blob/de2532f0a0d97067828dfd0077d0b852a845b920/experiments/BRAINSTORM.md> [49] <https://github.com/aswin-giridhar/london-ai-science-hackathon/blob/de2532f0a0d97067828dfd0077d0b852a845b920/splits/README.md> [50] <https://github.com/google-deepmind/science-skills/tree/68832757cbbf941c620b71df5756cf6e5cc287b0> [51] <https://github.com/google-deepmind/science->

[skills/blob/68832757cbbf941c620b71df5756cf6e5cc287b0/SKILL_LICENSES.md](#) [52]

<https://haematologica.org/article/view/5692> [53] <https://pmc.ncbi.nlm.nih.gov/articles/PMC3032881> [54]

<https://www.ebi.ac.uk/ipd/imgt/hla> [55] <https://github.com/ANHIG/IMGTHLA/blob/Latest/LICENCE.md>

18. Build completion checklist

These are future implementation acceptance criteria, not claims that training ran during this planning revision.

- [] Canonical data and frozen assignments pass schema, key, label, and complete two-edit cross-partition checks; diagnostic isolation/statistic gaps are resolved before relying on their outputs.
- [] B0/B1 and the matched frozen/adapted sequence controls produce versioned, real predictions using the declared selection protocol.
- [] L150 demonstrates actual adapter updates with the expected trainable scope and unchanged frozen original weights; any U150 run updates only its declared original layers.
- [] Embedding caches are invalidated when the encoder changes and rebuilt only for the frozen selected checkpoint.
- [] Verified constructs and Boltz probes produce correctly mapped real structures, or the structural task remains explicitly incomplete.
- [] G_conf/G_pair use identical eligible training rows, paired test inputs, the same frozen S* base, and the exact same sequence fallback when structure is unavailable.
- [] Final reporting includes the predeclared paired metrics, support, undefined-statistic handling, full/valid-cohort distinctions, failure counts, and measured compute cost.
- [] The demo displays actual outputs with correct predicted/reference/no-structure labels and no invented individual coverage guarantee or mechanistic interpretation.
- [] Optional unfreezing, contact-tensor, hydrogen-bond, ensemble, ProteinMPNN, replication, and calibration work is marked run/unrun honestly and does not displace the core deliverable.

A completed negative 3D experiment is a legitimate scientific result. An unrun 3D branch is not completion of the requested architecture, even if a sequence-only fallback is accepted as a narrower hackathon submission. No accuracy improvement is claimed until the locked evaluation supplies evidence.